# CR Circuit Response to Step, Pulse, and Square Inputs and Low Pass, High Pass Filter

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## 1 Step Input

Step signal is represented as:

$$u(t) = \begin{cases} 0, & t < 0 \\ 1, & t \geq 0 \end{cases}$$

**Circuit:**

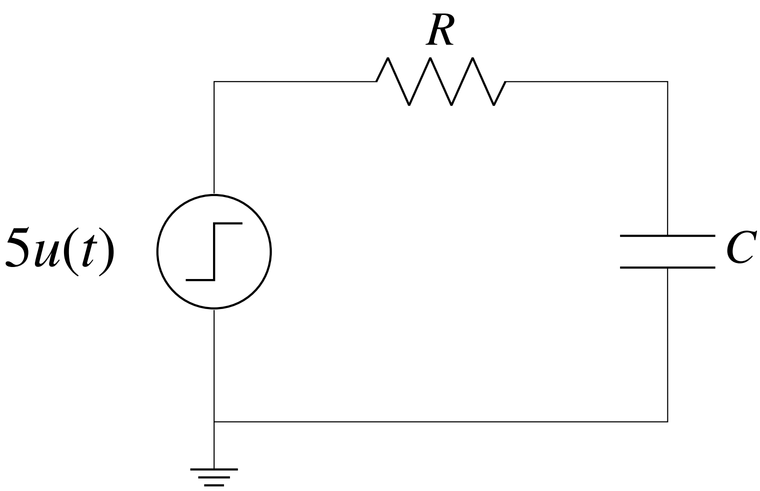


Figure 1: CR Circuit with Step Input

**Charging waveform**:
According to Kirchhoff's Voltage Law (KVL), the sum of voltages in the loop must equal the input voltage $V_{in}(t)$.
For the charging phase, the input is a step voltage $V_{in}(t) = V \cdot u(t)$

**Derivation:**

$$V_{in}(t) = V_C(t) + V_R(t) \tag{1}$$

For $t > 0$, $V_{in}(t) = V$.

$$V = \frac{1}{C}\int i(t)\,dt + i(t)R \tag{2}$$

Differentiating with respect to $t$:

$$0 = \frac{i(t)}{C} + R\frac{di(t)}{dt} \tag{3}$$

$$\implies \frac{di}{i} = -\frac{1}{RC}dt \tag{4}$$

$$\int_{I_0}^{i(t)} \frac{1}{i}\,di = \int_0^t -\frac{1}{RC}\,dt \tag{5}$$

$$\ln(i)\Big|_{I_0}^{i(t)} = -\frac{t}{RC} \tag{6}$$

$$\ln\left(\frac{i(t)}{I_0}\right) = -\frac{t}{RC} \tag{7}$$

$$\implies i(t) = I_0 e^{-\frac{t}{RC}} \tag{8}$$

At $t = 0$, the capacitor is uncharged and acts as a short circuit. Thus, $V_R(0) = V$, which means $I_0 = \frac{V}{R}$.

$$i(t) = \frac{V}{R}e^{-\frac{t}{RC}} \tag{9}$$

Since the output is across the resistor, $V_{out}(t) = V_R(t) = i(t)R$:

$$\mathbf{V_{out}(t) = Ve^{-\frac{t}{RC}}} \tag{10}$$

**Discharging (Step Down)**

$$\implies i(t) = I_0 e^{-\frac{t}{RC}} \tag{11}$$

At $t = 0$, the capacitor acts as a source of voltage $V$. However, the current now flows out of the capacitor and through the resistor in the opposite direction. Therefore, $V_R(0) = -V$, which means $I_0 = -\frac{V}{R}$.

$$i(t) = -\frac{V}{R}e^{-\frac{t}{RC}} \tag{12}$$

The output voltage is:

$$\mathbf{V_{out}(t) = -Ve^{-\frac{t}{RC}}} \tag{13}$$

**Conclusion:**

Here in the input the value of V = 5V and R = 820Ω and C = 1μF

$$\mathbf{V_{out}(t)} = \begin{cases} 5e^{-\frac{t}{RC}} \cdot u(t) & \text{for Step-Up (Charging)} \\ -5e^{-\frac{t}{RC}} \cdot u(t) & \text{for Step-Down (Discharging)} \end{cases} \tag{14}$$

## 1.1 Waveforms

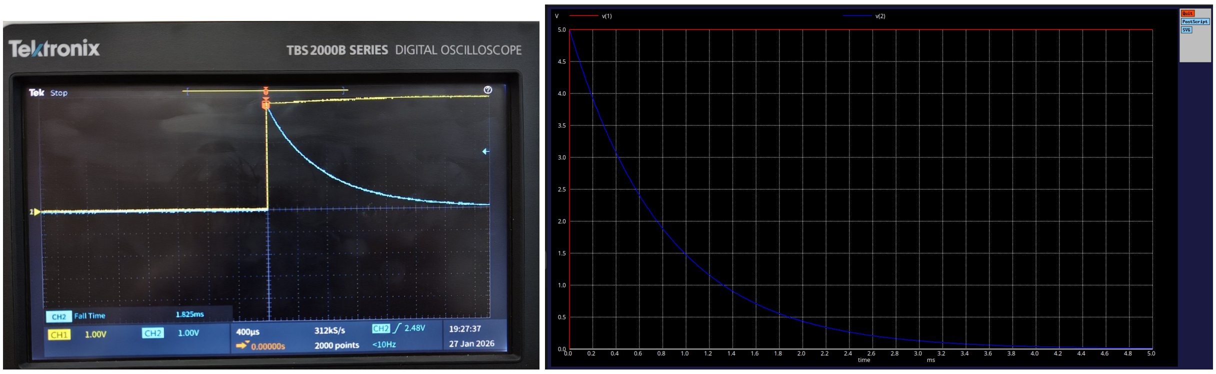


Figure 2: Charging Waveform of CR circuit verified with ngspice simulation

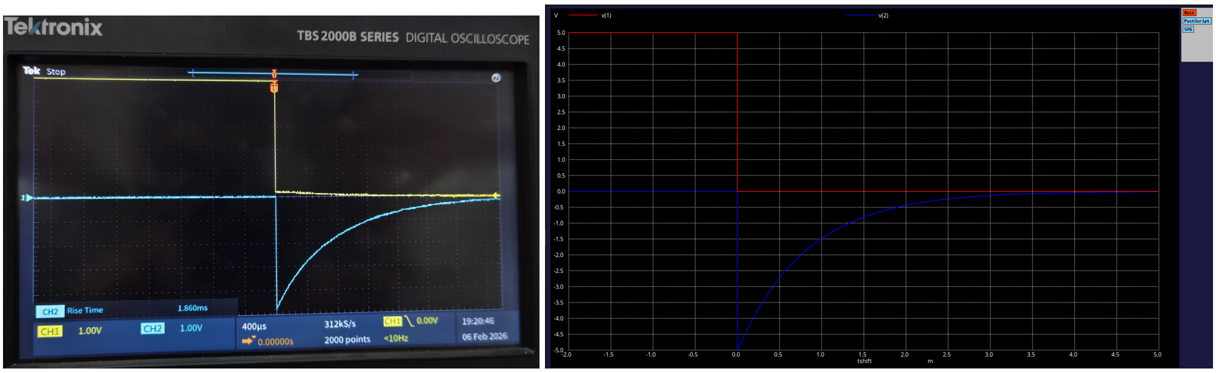


Figure 3: Discharging Waveform of RC circuit verified with ngspice simulation

## 1.2 Time constant calculation

$$V_R(t) = 5e^{-\frac{t}{RC}} \tag{15}$$

At t = RC

$$V_R(t) = 5e^{-1} \tag{16}$$

$$\implies V_R(t) = 5 \times 0.368 = 1.83V \tag{17}$$

i.e time to taken by the resistor to decrease it's voltage from the peak value to 36.8% of it's peak voltage is defined as Time constant($\tau$) of the circuit.
From the waveforms obtained in DSO we get the time constant as

$$\tau \approx 820\mu s \tag{18}$$

**Fall Time:**
The rise time ($t_r$) of an CR circuit is defined as the time required for the output pulse to decay from 90% of its peak value down to 10% of its peak value.

$$\implies \tau = \frac{t_\mathrm{r}}{2.2} \tag{19}$$

$$\implies \tau = \frac{1825}{2.2} \mu s \tag{20}$$

$$\tau \approx 830 \mu s \tag{21}$$

**Error:**

$$\text{Error} = \frac{830 - 820}{820} \times 100\% \tag{22}$$

$$\implies = \frac{10}{820} \times 100\% \tag{23}$$

$$\implies \text{Error} \approx 1.22\% \tag{24}$$

Experimental value of time constant almost matches the original value of time constant with an error <1.5%
**Conclusion:**

- In the case of a charging waveform for step input for t>0 the voltage across the resistor will exponentially decrease to the relation $V_R(t) = 5e^{-\frac{t}{RC}}$
- In the case of a discharging waveform for a step input the voltage across the resistor will exponentially increase from -5V to 0V according to the relation $V_R(t) = -5e^{-\frac{t}{RC}}$
- The above results match exactly with the ngspice simulations.

# 2 Pulse input

Pulse input is given by:

$$x(t) = \begin{cases} 0, & t < 0 \\ 5, & 0 \le t < T \\ 0, & t \ge T \end{cases}$$

**Circuit:**

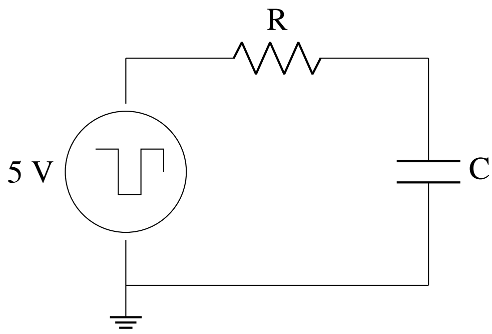


Figure 4: CR Circuit with Pulse Input

**Explanation:**

- From t = 0 to $t_{on}$ voltage of the resistor will decay according to the relation $V_R(t) = 5e^{-\frac{t}{RC}}$

- When the input $V_{in}$ transitions abruptly from 5V to 0V at $t = t_{on}$:

$$V_R(T_{on}) = V_{in}(T_{on}) - V_C(T_{on}^-) \tag{25}$$

$$V_R(T_{on}) = 0 - V_C \implies V_R = -V_C \tag{26}$$

- This creates a sharp jump into the negative region of the oscilloscope display.

- During the interval where $V_{in} = 0$, the capacitor discharges through the resistor. The discharge current $I(t)$ flows in the opposite direction to the charging current. The voltage of the resistor is governed by:

$$V_R(t) = -5 \cdot e^{-\frac{t}{RC}} \tag{27}$$

- As the time increases where $V_{in} = 0$ the voltage of the resistor tends to 0 according to (27)

## 2.1 Wave forms


Figure 5: CR Circuit with Pulse of with 20% duty cycle


Figure 6: CR Circuit with Pulse Input of 50% duty cycle

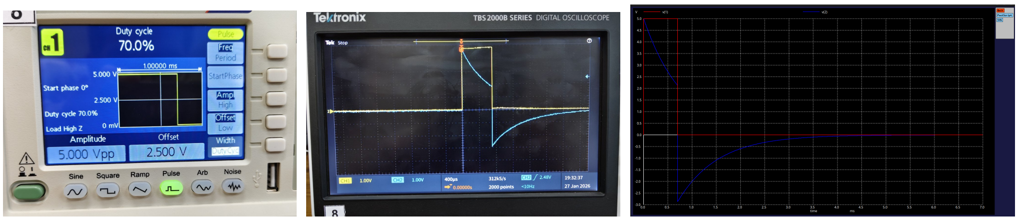


Figure 7: CR Circuit with Pulse Input of 70% duty cycle

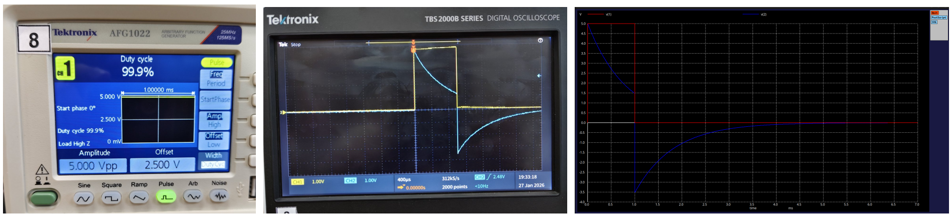


Figure 8: CR Circuit with Pulse Input of 99.9% duty cycle

**Comparision Table:**
Let t′ be the on time i.e the time where the voltage transition occurs from 5V to 0V

| **Duty Cycle**(%) | $\mathbf{V_R}$(t′-) | $\mathbf{V_R}$(t′+) |
|---|---|---|
| 20.0 | 3.916V | -1.1V |
| 50.0 | 2.72V | -2.28V |
| 70.0 | 2.13V | -2.87V |
| 99.9 | 1.48V | -3.52V |

**Conclusion:**

- As the duty cycle the voltage of the resistor at the transition time decreases
- After that the voltage of resistor starts from -$V_C(t)$ and gradually attains 0V as time progress

# 3 Square-Wave input

The square wave input voltage $v_{in}(t)$ with a period $T$ is defined as:

$$v_{in}(t) = \begin{cases} 5, & 0 \leq t < \frac{T}{2} \\ 0, & \frac{T}{2} \leq t < T \end{cases} \tag{28}$$

with the periodicity condition $v_{in}(t + T) = v_{in}(t)$.
**Circuit:**

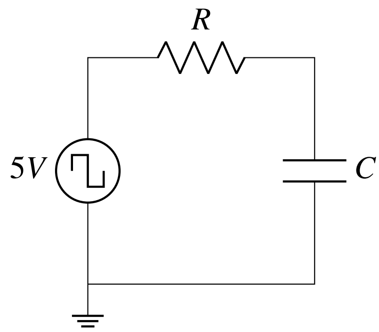


Figure 9: CR Circuit with a 5V Square Wave Input

## 3.1 CR>>T

In this case the value of $\tau = 820\mu$s so, let′s take the frequency of the square wave as 100kHz.Then, T = 10$\mu$s
**Calculation:**

- $V_R(t) = 5 \cdot e^{-\frac{t}{RC}} \implies V_R(t) = V_{in}(t) - V_c(t)$
- Capacitor takes $\approx$ 5time constants to chargei.e $\approx$ 4.1ms.
- The pulse width is very short (5$\mu$s). This is less than 1% of the time constant. During the ”ON” cycle, the capacitor has time to begin charging before the input switches ”OFF”.
- The capacitor cannot follow the rapid switching of the input. It settles at the point where the charge accumulating during the ”ON” phase $\approx$ equals the charge draining during the ”OFF” phase.
- In the **Steady State**, the capacitor reaches an average charge equal to the DC component of the input. For a 0V to 5V square wave, the average value is 2.5V.
- At $t(0^+)$, the input jumps from 0V to 5V. Since $V_C \approx 2.5$V, the output is:

$$V_R(0^+) = V_{in}(0^+) - V_C \approx 5\text{V} - 2.5\text{V} = 2.5\text{V} \tag{29}$$

- Similarly, when the input drops to 0V:

$$V_R(t_{fall}^{+}) = 0\text{V} - 2.5\text{V} = -2.5\text{V} \tag{30}$$

- The exponential decay seen at 1 kHz is now nearly invisible. So, it is difficult to capture transient response in this case .The steady state looks like a flat-top square wave centered at 0V.

**Waveforms**

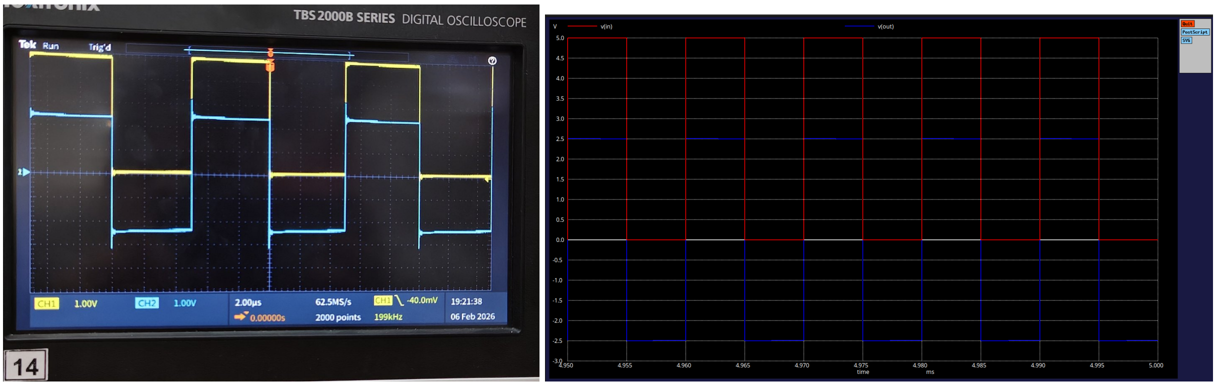


Figure 10: CR>>T case verified with ngspice simulation

## 3.2 RC $\approx$ T

The value of $\tau = 820\mu s$ so, let′s take the frequency of the square wave as 1kHz.Then, T = 1000$\mu$s.
For a square wave of time period T it′s value is 5v for $\frac{T}{2}$ and 0 for remaining time.
**Calculation:**

- At $t = 0$, the capacitor is uncharged ($V_C(0^-) = 0\,\text{V}$). Because $V_C$ cannot change instantaneously:

$$V_R(0^+) = V_{in}(0^+) - V_C(0^-) = 5\,\text{V} - 0\,\text{V} = 5\,\text{V} \tag{31}$$

- As $t$ approaches 0.5 ms,

$$V_R(t) = e^{-\frac{500}{820}} \tag{32}$$
$$\implies V_R(t) = 2.72V \text{ and } V_C(t) = 2.28V \tag{33}$$

- When the input drops to 0 V, the resistor voltage becomes:

$$V_R(0.5\,\text{ms}^+) = 0\,\text{V} - 2.28\,\text{V} = -2.28\,\text{V} \tag{34}$$

- from t = 0.5ms to t = 1ms $V_R(t) = 0 - V_C(t)$

$$V_R(t) = -2.28 \cdot e^{-\left(\frac{t-0.5}{RC}\right)} \tag{35}$$

$$\implies V_C(1\,\text{ms}^-) = 2.28 \times e^{-0.61} \tag{36}$$

$$\implies V_C(1\,\text{ms}^-) = 1.24V \tag{37}$$

- when t > 1ms $V_C(1\,\text{ms}^+) = V_C(1\,\text{ms}^-) = 1.24V$

$$V_R(t) = 3.76V \tag{38}$$

- At final, in steady state the output settles at ±3.24V.

**Waveforms**

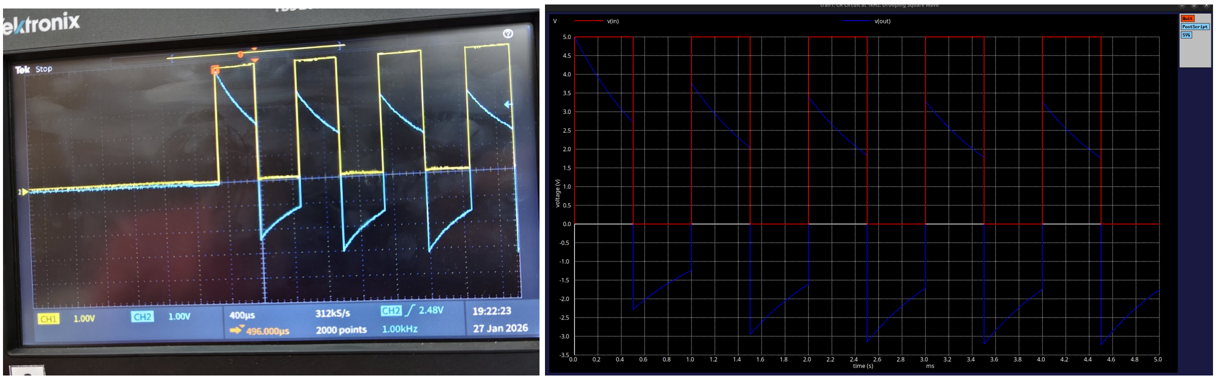


Figure 11: CR≈T case Transient response verified with ngspice simulation

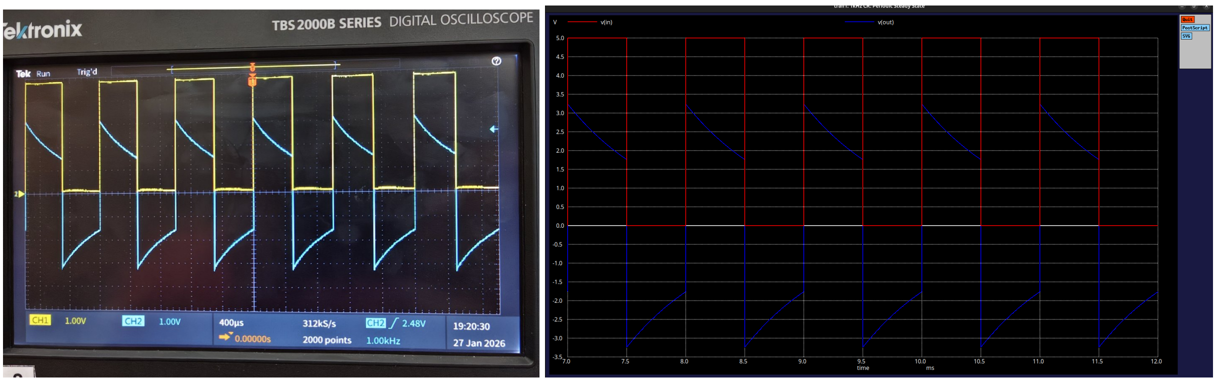


Figure 12: CR≈T case Steady state verified with ngspice simulation

## 3.3 RC<<T

In this case the value of $\tau = 820\mu$s so, let′s take the frequency of the square wave as 1Hz.Then, T = 1s.

**Calculation:**

- Capacitor takes $\approx 5\tau$ to charge completely so it takes around 4.1ms to charge completely but the on time of the square wave is veery high i.e 0.5s.
- So Capacitor takes <1% of the on time completely so most of the time capacitor acts as open circuit.so,$V_R(t) = V_{\text{in}} - V_C(t) = 0$ when capacitor acts as open circuit.
- When the input square wave is at a steady $5V$ or $0V$, the capacitor acts as an open circuit ($I = 0$). By Ohm's Law, $V_R = IR = 0V$. This results in the long horizontal segments observed on the $0V$ axis in the DSO and simulation traces.
- The voltage of the resistor is proportional to current through it.

$$V_R(t) = R \cdot i(t) \tag{39}$$

$$\implies V_R(t) = RC\frac{dV_C}{dt} \tag{40}$$

$$V_R(t) \approx RC\frac{dV_{in}}{dt} \tag{41}$$

  At the transition edges of a square wave, the derivative $\frac{dV_{in}}{dt}$ is a Dirac delta-like pulse.

- Generally the dirac delta function is represented as

$$\delta(t) = \begin{cases} +\infty, & t = 0 \\ 0, & t \neq 0 \end{cases} \tag{42}$$

- The rising edge produces a positive impulse $+\delta(t)$, while the falling edge produces a negative impulse $-\delta(t)$, explaining the alternating upward and downward "needles" seen in the steady-state graph.

**Wave forms:**

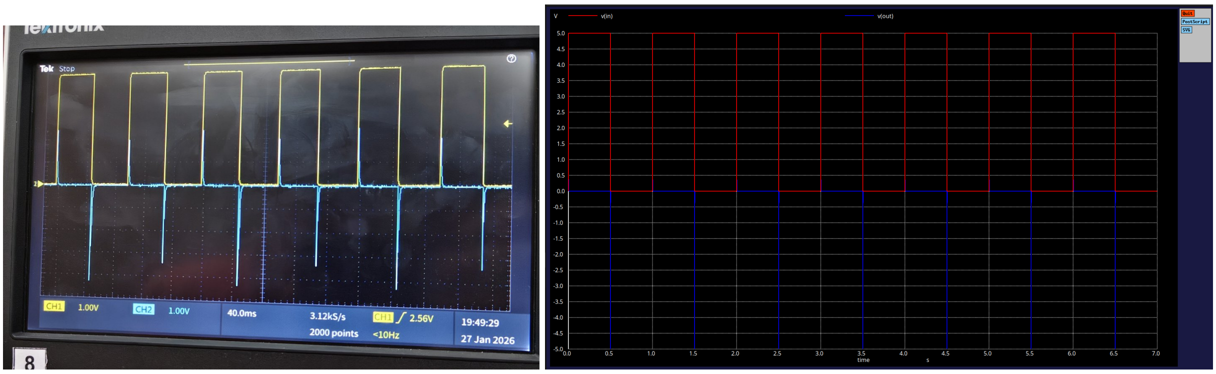


Figure 13: CR<<T case Steady state verified with ngspice simulation

# 4 Low Pass Filter

- **Definition:** A low pass filter circuit is a filter that passes signals with frequencies lower than cutoff frequency ($f_c$) while attenuating (reducing the amplitude of) signals with frequencies higher than that cutoff.
- **Cut−off frequency:** The point where the output signal amplitude falls to 70.7% (−3 dB) of the input signal voltage.
- In a standarad RC Low-Pass Filter, the output voltage ($V_{out}$) is measured across the capacitor.

## 4.1 Calculations

**Gain:**

- gain = $\frac{|V_{\text{out}}|}{|V_{\text{in}}|}$
- The impedances of the resistor and capacitor are $Z_R = R$ and $Z_C = \frac{1}{j\omega C}$, respectively. The transfer function is given by:

$$H(\omega) = \frac{V_{out}}{V_{in}} = \frac{Z_C}{Z_R + Z_C} \tag{43}$$

Substituting the impedance values:

$$H(\omega) = \frac{\frac{1}{j\omega C}}{R + \frac{1}{j\omega C}} = \frac{1}{1 + j\omega RC} \tag{44}$$

$$\implies \text{gain} = |H(\omega)| = \sqrt{H(\omega) \cdot H^*(\omega)} = \frac{1}{\sqrt{1^2 + (\omega RC)^2}} \tag{45}$$

- The cut-off frequency is defined as the point where the gain magnitude is $1/\sqrt{2}$ of the maximum gain. Setting the equation (45) equal to this value:

$$\frac{1}{\sqrt{1+(\omega_c RC)^2}} = \frac{1}{\sqrt{2}} \tag{46}$$

  Squaring both sides and solving for $\omega_c$:

$$1+(\omega_c RC)^2 = 2 \implies (\omega_c RC)^2 = 1 \implies \omega_c = \frac{1}{RC} \tag{47}$$

$$2\pi f_c = \frac{1}{RC} \implies f_c = \frac{1}{2\pi RC} \tag{48}$$

- Equation 45 can be written using cutoff frequency as $\omega_c = \frac{1}{RC}$,

$$|H(\omega)| = \frac{1}{\sqrt{1+\left(\frac{\omega}{\omega_c}\right)^2}} \tag{49}$$

## 4.2 Bode plots

**Magnitude response (dB) vs frequency on log scale :**

- **Y-axis:** The values on the y-axis are represented by the power gain in decibels, calculated as:

$$Y_{dB} = 20 \cdot \log_{10}(\text{Gain}) \tag{50}$$

- **X-axis:** While the actual frequency values ($f$) are labeled on the x-axis, the physical distance between them is measured on a **logarithmic scale**.
- **Scale Characteristic:** This means that the distance between any two decades is constant. For example, the physical distance between $10^2$ Hz and $10^3$ Hz is exactly the same as the distance between $10^3$ Hz and $10^4$ Hz.

**Phase response vs frequency on log scale :**

- **Y-axis:** The values on the y-axis are the phase difference between the waves of output and input measured in degrees.
- **X-axis:** While the actual frequency values ($f$) are labeled on the x-axis, the physical distance between them is measured on a **logarithmic scale**.
- From (44) The phase shift $\phi$ introduced by the filter is the argument of the transfer function:

$$\phi(\omega) = \angle H(\omega) = -\tan^{-1}(\omega RC) = -\tan^{-1}\left(\frac{\omega}{\omega_c}\right) \tag{51}$$

- when $\omega << \omega_c$ phase difference between them is nearly 0°..At the cutoff frequency ($\omega = \omega_c$), the phase shift is $-45°$.At $\omega >> \omega_c$ phase difference between them is ≈ -90°.

**Note:**

- In DSO we measured phase difference with respect to ch1 i.e ch1 - ch2 is the measured value on DSO.But, the required one is ch2 - ch1 So negative of phase difference observed is plotted
- Mathematically according to the calculations the minimum phase difference is ≈ -90°.But, actually in DSO when $\omega >> \omega_c$ the output is almost not detected and the phase difference observed in the DSO will be < -90°
- So, in DSO if phase difference $< -90°$ the phase difference can be taken as ≈ -90°

## 4.3 Wave forms

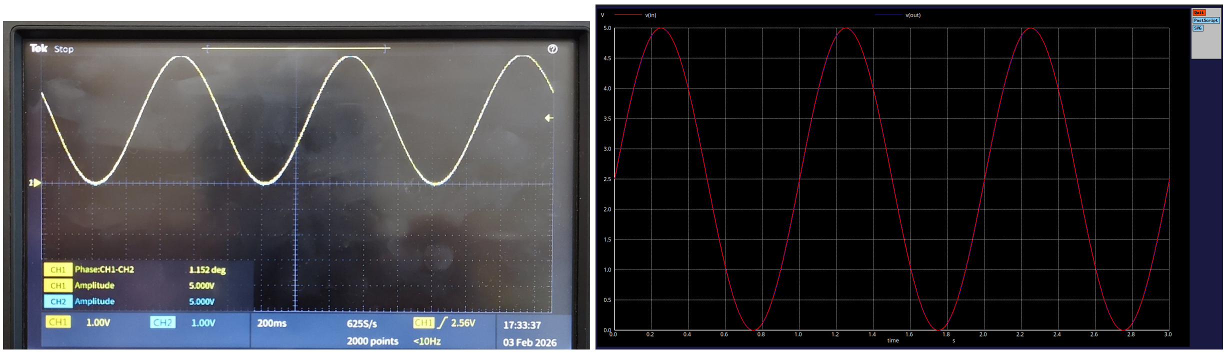


Figure 14: Low pass filter with frequency 1Hz

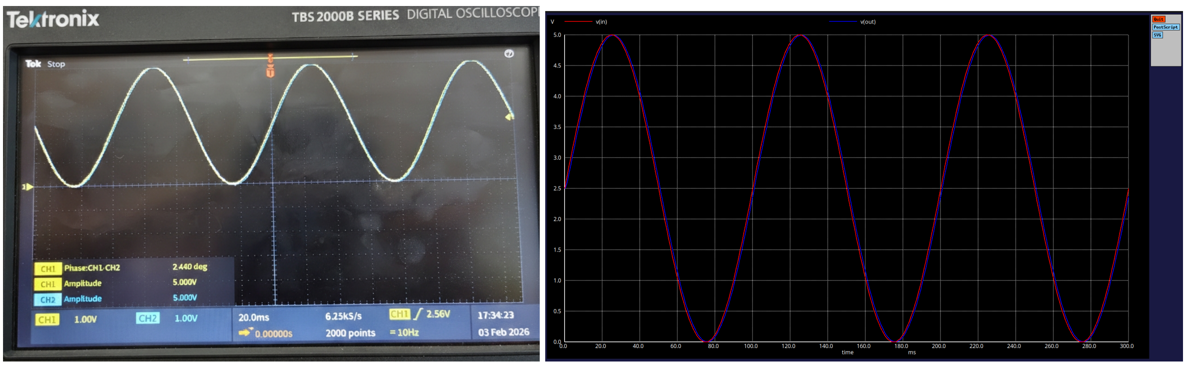


Figure 15: Low pass filter with frequency 10Hz

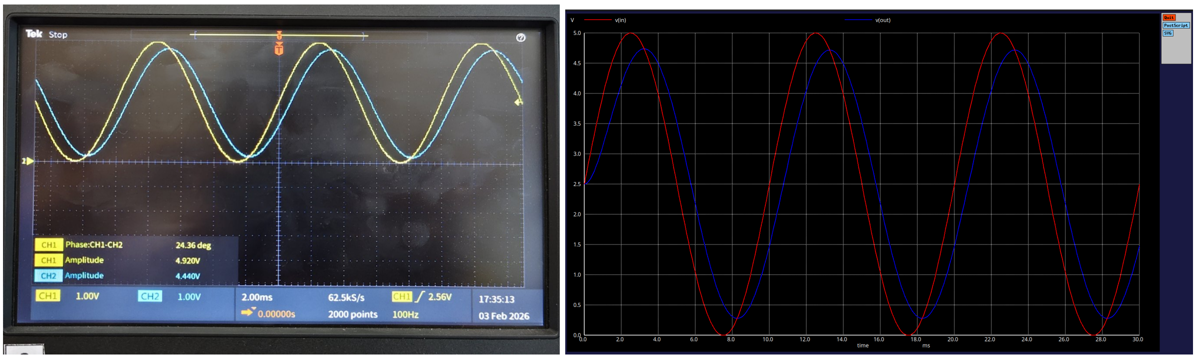


Figure 16: Low pass filter with frequency 100Hz

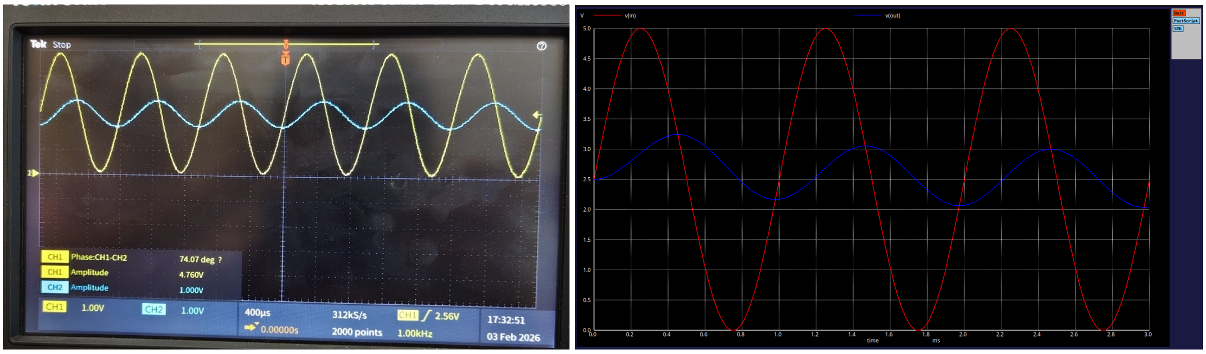


Figure 17: Low pass filter with frequency 1KHz

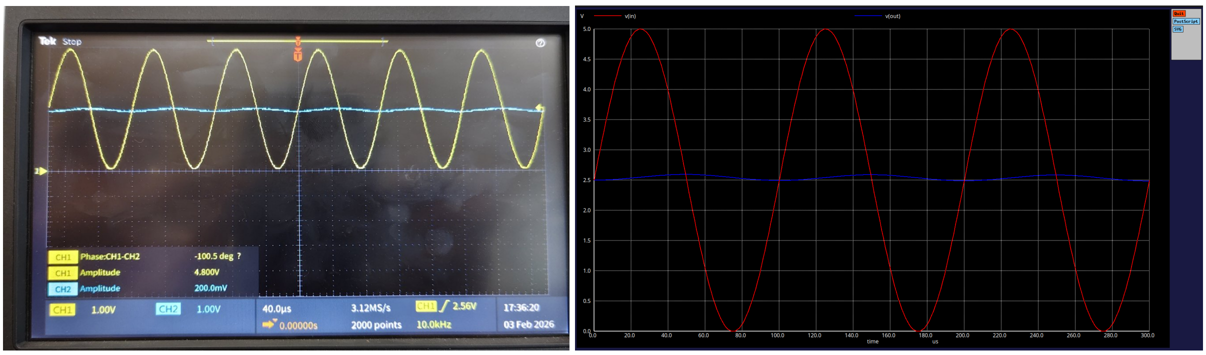


Figure 18: Low pass filter with frequency 10KHz

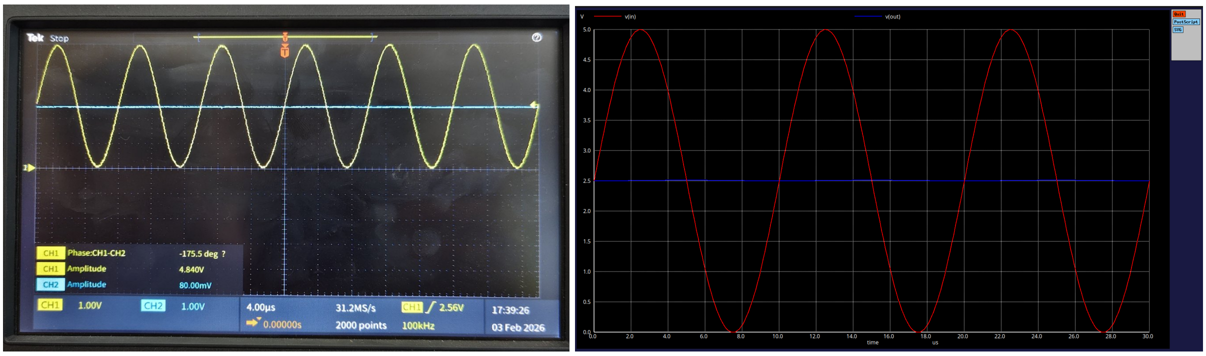


Figure 19: Low pass filter with frequency 100KHz

**Cut−off frequency:**
Cut−off frequency for thr given circuit(R = 820Ω and C = 1µF) given by

$$f_c = \frac{1}{2\pi RC} \tag{52}$$

$$\implies f_c = \frac{1}{2\pi \times 820 \times 10^3 \times 1 \times 10^{-}6} \tag{53}$$

$$\implies f_c = 194.09Hz \tag{54}$$

| Frequency (Hz) | Linear Gain $\|H(f)\|$ | Output Voltage $V_{out}$ (V) | Gain (dB) |
|---|---|---|---|
| 1 | 0.9999 | 5.000 | -0.0001 |
| 10 | 0.9987 | 4.993 | -0.0115 |
| 100 | 0.8889 | 4.445 | -1.0225 |
| $10^3$ | 0.1905 | 0.953 | -14.4005 |
| $10^4$ | 0.0194 | 0.097 | -34.2415 |
| $10^5$ | 0.0019 | 0.010 | -54.2399 |

Table 1: Gain at different frequencies($R = 820\,\Omega, C = 1\,\mu\text{F}, V_{in} = 5\text{V}$)

The above Theoretical values almost matches the experimental data

**Note:**
The experimental input amplitude measured was $< 5.0$ V in some cases. This error can be attributed to the following factors:

- The 50 Ω internal source impedance of the function generator creates a voltage divider with the circuit's input impedance, slightly reducing the voltage delivered to the nodes.
- The DSO vertical resolution at 1 V/div may introduce a small measurement offset during the analog-to-digital conversion process.

## 4.4 Graphs:

**Magnitude Response**

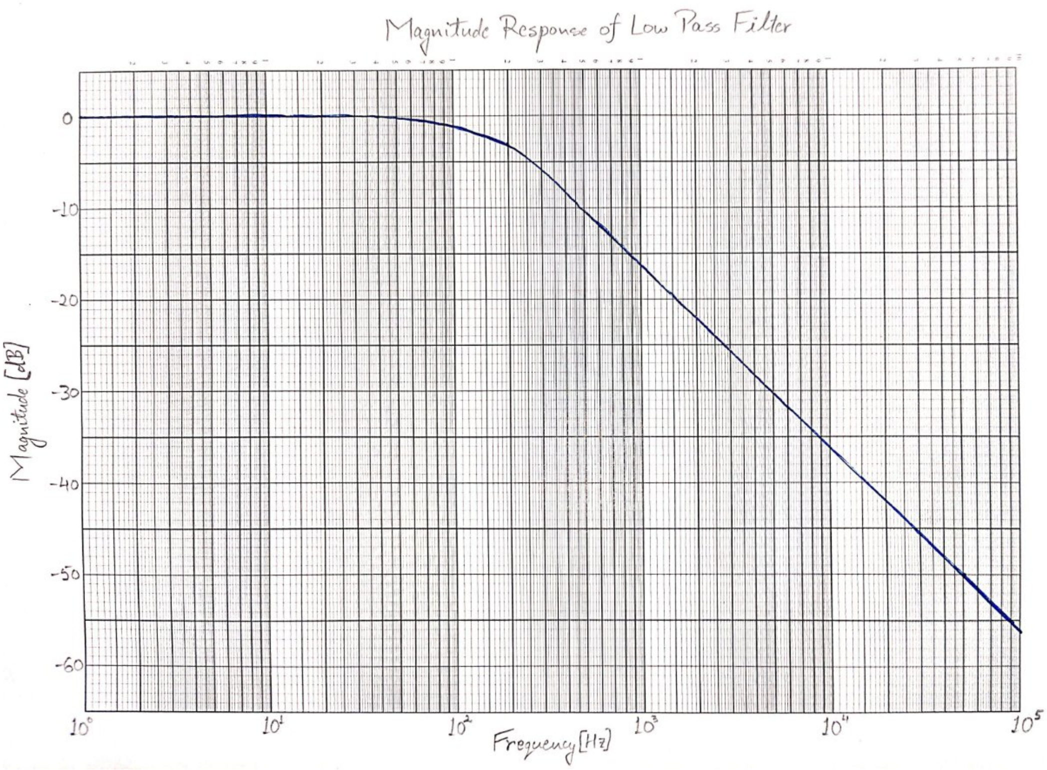


Figure 20: Experimental Plot of a High pass filter

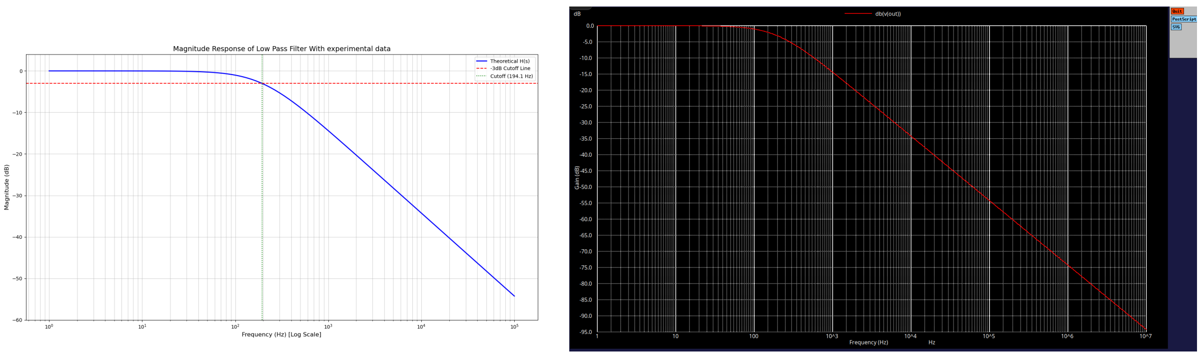


Figure 21: Python plot and ngspice plot of a Low pass filter

Experimental graph plotted matches exactly with the python plot and ngspice plot

**Phase Response**

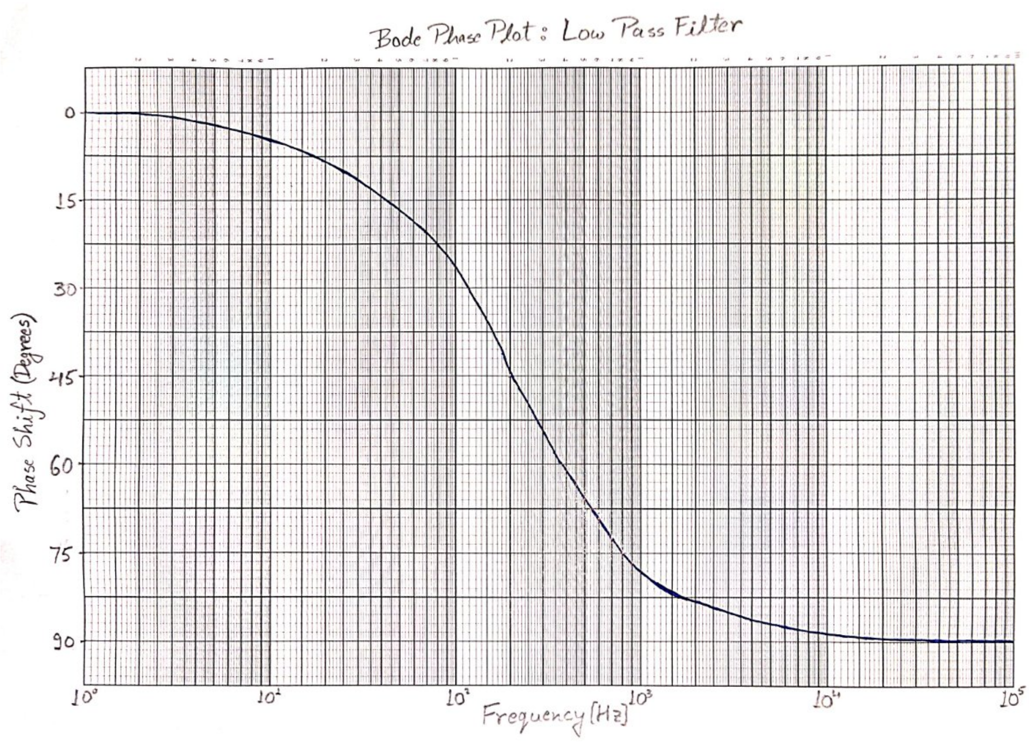


Figure 22: Experimental Plot of a Low pass filter

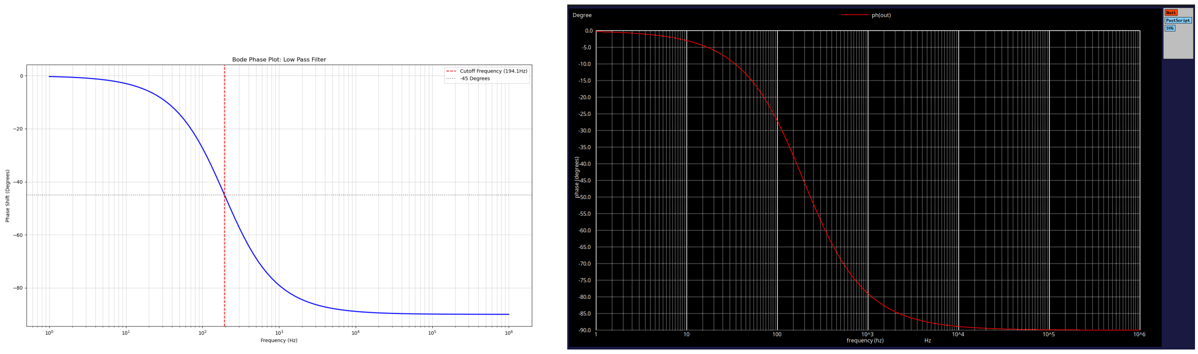

Figure 23: Python plot and ngspice plot of a Low pass filter

Above Experimental graph of phase response of Low pass filter matches exactly with the python plot and ngspice plot

**Slope**

- $\omega < \omega_c$, the slope is approximately 0 dB/decade.Means the signal passes through with almost no loss in amplitude.

- $\omega = \omega_c$ This is the point of the curve where the slope begins to transition.
- $\omega > \omega_c$ The slope becomes a constant -20 dB/decade.

# 5 High Pass Filter

- **Definition:** A High pass filter circuit is a filter that passes signals with frequencies higher than cutoff frequency ($f_c$) while attenuating (reducing the amplitude of) signals with frequencies lower than that cutoff.
- **Cut−off frequency:** The point where the output signal amplitude falls to 70.7% (−3 dB) of the input signal voltage.
- In a standarad RC High-Pass Filter, the output voltage ($V_{out}$) is measured across the resistor.

## 5.1 Calculations

**Gain:**

- gain = $\frac{|V_{out}|}{|V_{in}|}$
- The impedances of the resistor and capacitor are $Z_R = R$ and $Z_C = \frac{1}{j\omega C}$, respectively. For a high-pass filter, the output is taken across the resistor. The transfer function is given by:

$$H(\omega) = \frac{V_{out}}{V_{in}} = \frac{Z_R}{Z_R + Z_C} \tag{55}$$

Substituting the impedance values:

$$H(\omega) = \frac{R}{R + \frac{1}{j\omega C}} = \frac{j\omega RC}{1 + j\omega RC} \tag{56}$$

$$\implies \text{gain} = |H(\omega)| = \frac{\omega RC}{\sqrt{1 + (\omega RC)^2}} \tag{57}$$

- The cut-off frequency is defined as the point where the gain magnitude is $1/\sqrt{2}$ of the maximum gain. Setting equation (57) equal to this value:

$$\frac{\omega_c RC}{\sqrt{1 + (\omega_c RC)^2}} = \frac{1}{\sqrt{2}} \tag{58}$$

Squaring both sides and solving for $\omega_c$:

$$\frac{(\omega_c RC)^2}{1+(\omega_c RC)^2} = \frac{1}{2} \implies (\omega_c RC)^2 = 1 \implies \omega_c = \frac{1}{RC} \tag{59}$$

$$2\pi f_c = \frac{1}{RC} \implies f_c = \frac{1}{2\pi RC} \tag{60}$$

- Equation 57 can be written using cutoff frequency as $\omega_c = \frac{1}{RC}$. Dividing numerator and denominator by $\omega RC$:

$$|H(\omega)| = \frac{1}{\sqrt{1+\left(\frac{\omega_c}{\omega}\right)^2}} \tag{61}$$

## 5.2 Bode Plots

**Magnitude response (dB) vs frequency on log scale :**

- **Y-axis:** The values on the y-axis are represented by the power gain in decibels, calculated as:

$$Y_{dB} = 20 \cdot \log_{10}(\text{Gain}) \tag{62}$$

- **X-axis:** While the actual frequency values ($f$) are labeled on the x-axis, the physical distance between them is measured on a **logarithmic scale**.
- **Scale Characteristic:** This means that the distance between any two decades is constant. For example, the physical distance between $10^2$ Hz and $10^3$ Hz is exactly the same as the distance between $10^3$ Hz and $10^4$ Hz.

**Phase response vs frequency on log scale :**

- **Y-axis:** The values on the y-axis are the phase difference between the waves of output and input measured in degrees.
- **X-axis:** While the actual frequency values ($f$) are labeled on the x-axis, the physical distance between them is measured on a **logarithmic scale**.
- From (56) The phase shift $\phi$ introduced by the filter is the argument of the transfer function:

$$H(\omega) = \frac{j\omega RC}{1+j\omega RC} \tag{63}$$

$$\phi(\omega) = \angle H(\omega) = \frac{\pi}{2} - \tan^{-1}(\omega RC) = \tan^{-1}\left(\frac{1}{\omega RC}\right) = \tan^{-1}\left(\frac{\omega_c}{\omega}\right) \tag{64}$$

- when $\omega << \omega_c$ phase difference between them is$\approx 90°$..At the cutoff frequency ($\omega = \omega_c$), the phase shift is $45°$.At $\omega >> \omega_c$ phase difference between them is $\approx 0°$.

**Note:**

- In DSO we measured phase difference with respect to ch1 i.e ch1 - ch2 is the measured value on DSO.But, the required one is ch2 - ch1 So negative of phase difference observed is plotted
- Mathematically according to the calculations the maximum phase difference is $\approx 90°$.But, actually in DSO when $\omega << \omega_c$ the output is almost not detected and the phase difference will go beyond $90°$
- So, in DSO if phase difference $> 90°$ the phase difference can be taken as $\approx 90°$

## 5.3 Waveforms

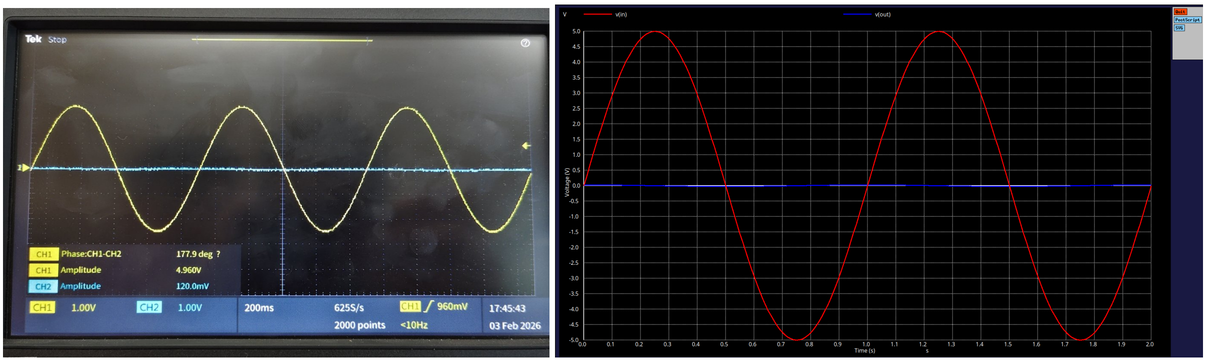


Figure 24: High pass filter with frequency 1Hz

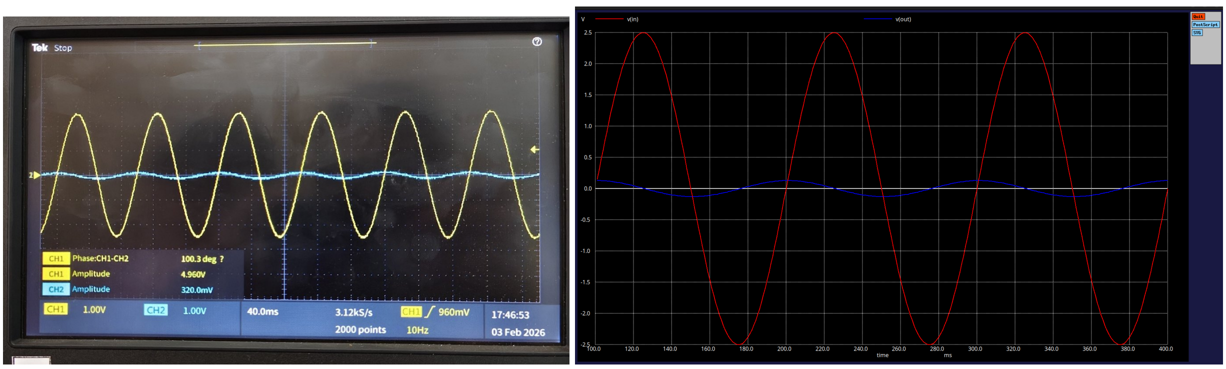


Figure 25: High pass filter with frequency 10Hz

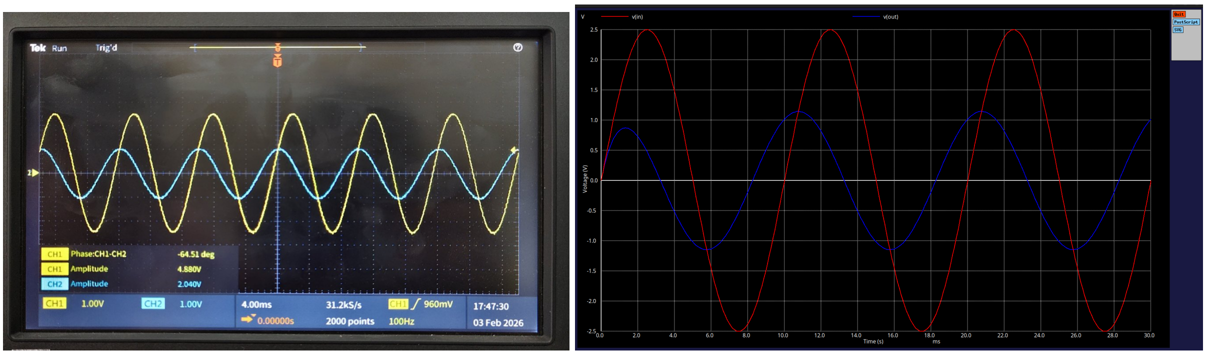


Figure 26: High pass filter with frequency 100Hz

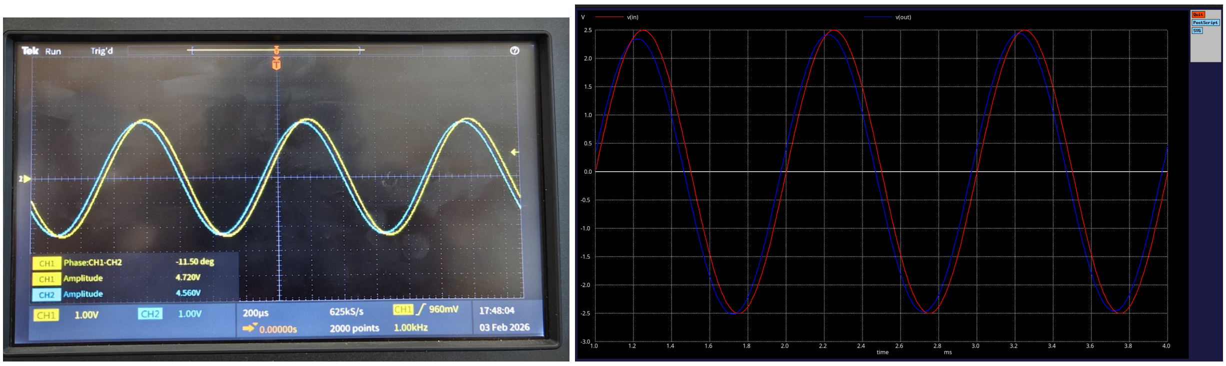


Figure 27: High pass filter with frequency 1kHz


Figure 28: High pass filter with frequency 10kHz

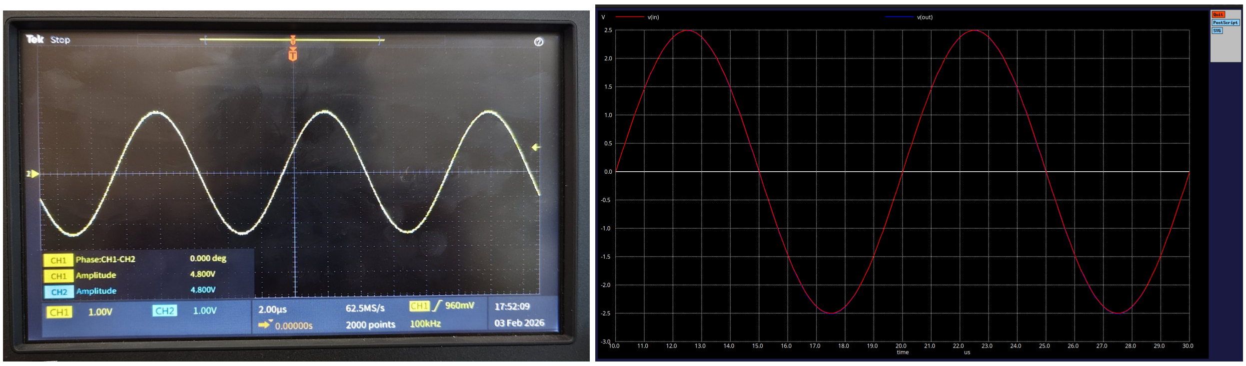


Figure 29: High pass filter with frequency 100kHz

## 5.4 Graphs:

### Magnitude Response

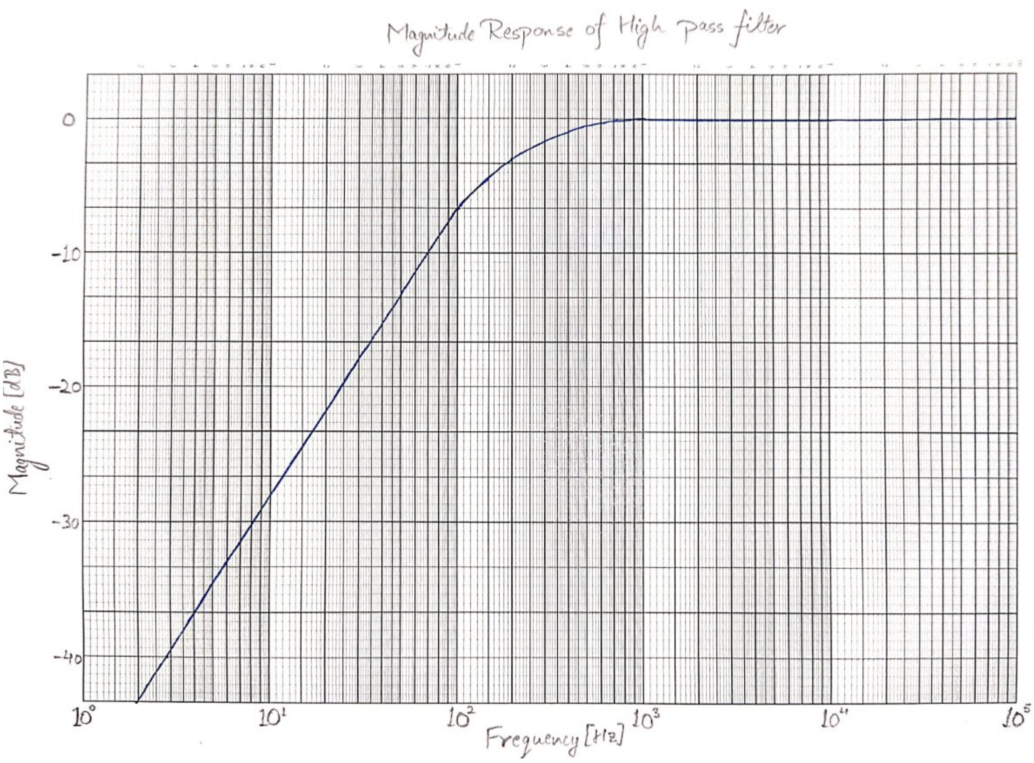


Figure 30: Experimental Plot of a High pass filter

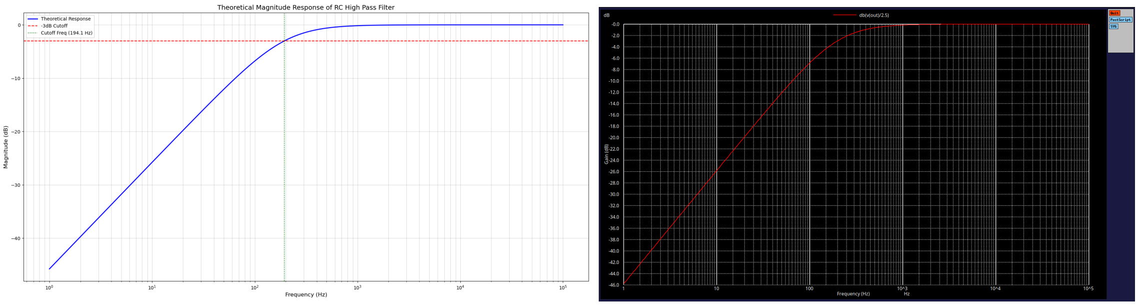


Figure 31: Python plot and ngspice plot of a High pass filter

## Phase Response

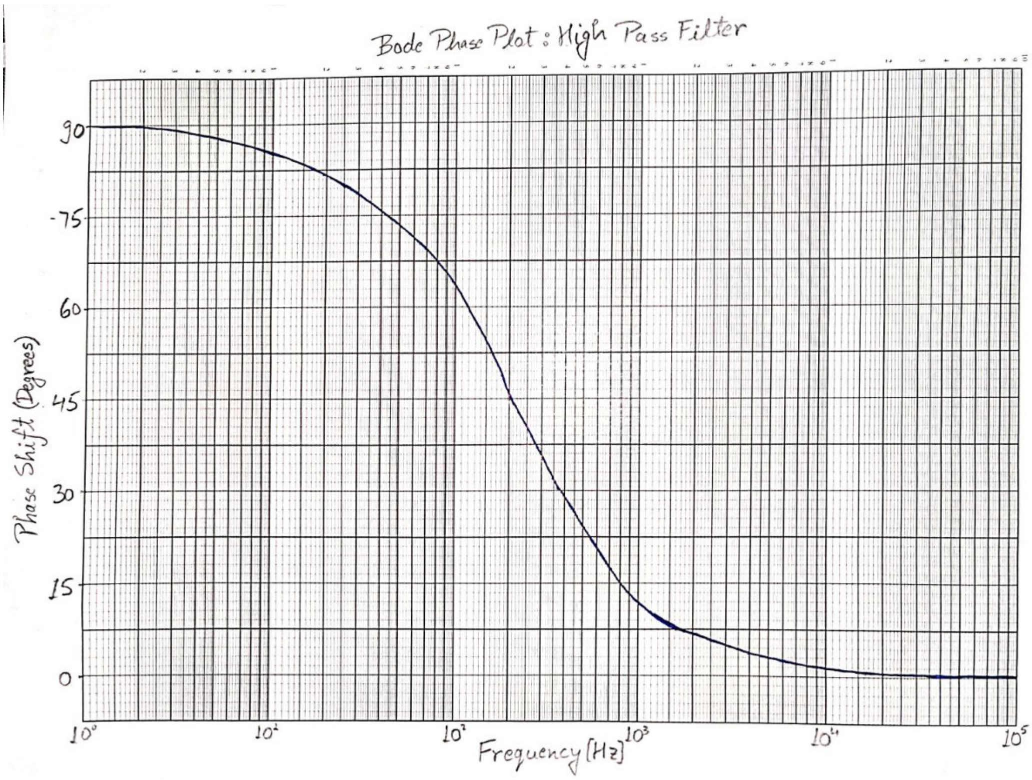


Figure 32: Experimental Plot of a High pass filter

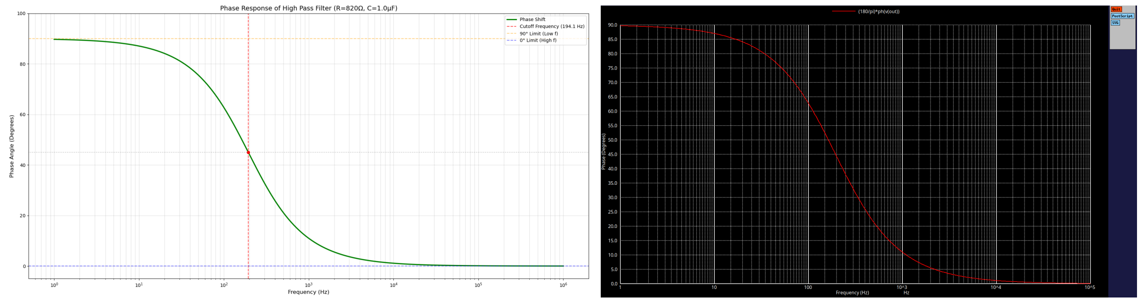

Figure 33: Python plot and ngspice plot of a High pass filter for phase response

Above Experimental graph of phase response of High pass filter matches exactly with the python plot and ngspice plot

**Cut−off frequency:**

Cut−off frequency for thr given circuit(R = 820Ω and C = 1µF) given by

$$f_c = \frac{1}{2\pi RC} \tag{65}$$

$$\implies f_c = \frac{1}{2\pi \times 820 \times 10^3 \times 1 \times 10-6} \tag{66}$$

$$\implies f_c = 194.09Hz \tag{67}$$

**Slope:**

- $\omega < \omega_c$, the slope is a constant +20 dB/decade. Means the signal is attenuated at lower frequencies.
- $\omega = \omega_c$ This is the point of the curve where the slope begins to transition.
- $\omega > \omega_c$, the slope is approximately 0 dB/decade. Means the signal passes through with almost no loss in amplitude.

| **Frequency (Hz)** | **Linear Gain $\lvert H(f)\rvert$** | **Output Voltage $V_{out}$ (V)** | **Gain (dB)** |
|---|---|---|---|
| 1 | 0.0051 | 0.0257 | -45.7600 |
| 10 | 0.0514 | 0.2573 | -25.7700 |
| 100 | 0.4579 | 2.2895 | -6.7840 |
| $10^3$ | 0.9816 | 4.9084 | -0.1610 |
| $10^4$ | 0.9998 | 4.9990 | -0.0010 |
| $10^5$ | 1.0000 | 5.0000 | 0.0000 |

Table 2: Gain at different frequencies ($R = 820\,\Omega, C = 1\,\mu$F, $V_{in} = 5$V)